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ESTIMATING A PRUDENT LEVEL OF PUBLIC DEBT IN CHILE

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Abstract

Fiscal sustainability is fundamental for macroeconomic stability and the economic development of countries. The increasing accumulation of public debt and the decline in economic growth in Chile have renewed focus on this issue. To address this problem, one alternative is a *dual* fiscal rule that incorporates an additional debt ceiling rule into the structural balance rule. This paper estimates a prudent debt level at which to set this ceiling using the methodology proposed by Eyraud et al. (2018). In this approach, probabilistic trajectories of fiscal variables are simulated, and the prudent level is defined as the maximum initial debt level compatible with trajectories in which the maximum feasible primary balance is not exceeded with high probability. The estimation for Chile yields a prudent level of 51% of GDP, though this result is sensitive to the calibration of the maximum feasible primary balance, which is not directly observable for Chile.

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I. Introduction

Fiscal sustainability is a fundamental pillar for macroeconomic stability, and therefore, for the economic and social development of countries. The increasing accumulation of public debt in Chile in recent years, coupled with a reduction in the level of economic growth, has renewed focus on this issue. One alternative to address this problem would be to establish a *dual* fiscal rule that incorporates an additional debt ceiling rule into the structural balance rule. This paper estimates a prudent debt level at which to set this debt ceiling through the maximum feasible primary balance methodology proposed in Eyraud et al. (2018). In this approach, probabilistic trajectories of debt and the primary balance as a percentage of GDP are simulated, incorporating shocks to the main variables affecting fiscal policy. The prudent level is defined as the maximum initial debt level that is compatible with trajectories in which the maximum feasible primary balance (MFPB) is not exceeded with high probability. The calibration for Chile yields a prudent debt level of approximately 51% of GDP.

Macroeconomic stability is not a product of chance but rather the result of implementing sound policies to prevent and address imbalances responsibly. When fiscal policy becomes an instrument for macroeconomic stability instead of a source of instability, the future capacity to respond to negative shocks using countercyclical policies is expanded, thereby reducing macroeconomic vulnerability. Furthermore, sustainability limits the financial vulnerability associated with the possibility of falling into a sovereign debt crisis, along with all the severe consequences that such an event may entail. Crises hinder the development of long-term projects and force a short-term focus, impeding investments and productive advancements.

In 1990, Chile had a debt level exceeding 50% of GDP, which, after an impressive reduction by 2007, reached a minimum of 3.8% of GDP. However, since then, the debt level has been steadily increasing to its current position of 38.2%. This has been the consequence of repeated fiscal deficits and lower GDP expansion. Adjusting for the effects of the economic cycle, primary balances have averaged -1.65% from 2007 to 2021 and -3.47% over the last 5 years, with the impact of the Covid-19 pandemic being particularly significant. In addition, interest expenses have been rising; in 2008, these represented 1.9% of tax revenue, whereas in 2020 they reached 4.5% of that year's revenue (International Monetary Fund, 2022). This implies either taking on even more debt or spending a larger portion of revenue on paying interest rather than on expenditures. This increased fiscal pressure has also caused the main rating agencies to progressively downgrade Chile's credit rating. Although Chile's public debt-to-GDP ratio remains low globally and in comparison to countries with similar credit ratings, it is essential to act cautiously, as a sovereign default scenario must be avoided above all else.

Fiscal rules are relatively new policy instruments. In 1992, after the European Community established a maximum debt level of 60% and a fiscal deficit limit of 3%, their use spread to other

countries. Following recommendations from the IMF, a growing number of developing and low-income countries have been adopting these types of rules. To date, they are present in 105 countries; 85 countries have fiscal debt rules, and in 75 of these, they are combined with budget balance rules (Davoodi et al., 2022). The rationale behind these rules stems from combating problems of moral hazard (authorities knowing that their excessive spending will be paid for by future generations), time inconsistency (authorities committing to certain spending levels to secure favorable terms for their debt, but later failing to comply), and general short-sightedness.

In particular, a debt rule would be justified because reaching a high level of debt is highly detrimental; in addition to the macroeconomic and financial vulnerability described, there is evidence that above a certain level, debt begins to act as a drag on economic growth, potentially reducing it (Cecchetti et al., 2011; Cottarelli and Jaramillo, 2012). On the other hand, it could delay politically costly but necessary structural reforms to increase revenue, and also alter the composition of expenditures, since higher interest payments could force reductions in other necessary spending. However, there would also be a trade-off in setting an excessively restrictive maximum debt level. First, the issuance of debt allows for the smoothing of economic cycle effects through public spending, which is valued by economic agents and is precisely what fiscal rules seek to achieve. Second, there are investments in physical capital (such as public works) or human capital (such as health and education) that are priorities and could be sustainably financed with debt, as they would be recovered through higher productivity translating into greater future revenue.

When the debt ceiling level is set appropriately and there is credibility that the authorities will respect it, the risk premium and macroeconomic volatility can be diminished because expectations regarding the management of current and future fiscal policy improve, benefiting the country. These types of rules aim for *convergence* to a sustainable level of public debt and commit governments to act sustainably in the medium term, which is incorporated into the expectations of agents. Furthermore, a fiscal rule based on debt-to-GDP levels (whether gross or net debt) would complement the structural balance rule, which was designed to achieve acyclical planning of fiscal policy, eliminating transient effects on revenues and expenditures generated by the short-term fluctuations inherent in the economic cycle. A debt ceiling is relatively simple to communicate to the public, helping to anchor expectations, generate pressure on policymakers for effective compliance, and signal how much fiscal space (the difference between the debt ceiling and the current level) exists.

However, it is also essential to highlight the problems that setting a debt ceiling can generate. First, authorities might take advantage by setting an excessively high level; by being opaque in its estimation and assumptions; and, consequently, by establishing a limit that is not truly restrictive. Second, if the debt ceiling is not communicated clearly, it could be interpreted instead as a debt anchor towards which the government must transition, which would generate pro-cyclical fiscal policy in the short term until that level is reached (Consejo Fiscal Autónomo, 2021). Lastly, this level might be appropriate for the initial situation, but after being applied for years, it could become too rigid

for new macroeconomic conditions, akin to the European Community’s stability pact which saw subsequent modifications in 2005 and 2011. In all these respects, the institutional framework within which these policies are embedded is fundamental for their successful implementation. Particularly important is the counterweight provided by fiscal councils in contributing to fiscal sustainability by evaluating fiscal policy management and alerting policymakers and citizens to potential problems.

A prudent debt level is defined as “A level of the debt-to-GDP ratio that ensures the sustainability of public finances with a high probability” (Consejo Fiscal Autónomo, 2021). In Eyraud et al. (2018), two methodologies are proposed to estimate it. In the first, the sustainability of public finances would be formulated as the maximum level of debt that ensures, over a given horizon, the *debt limit level* is not exceeded, with a certain probability (traditionally 5%). Such a level would reflect the point above which a sovereign default would occur. In the second (the one developed in this paper), it is calculated using an approach less frequent in the literature, following the idea that at high debt levels governments tend to respond with high primary balances to return to prudent levels. However, above a certain level, the balances necessary to stabilize the debt would not be realistic (according to historical evidence) because they exceed the *maximum feasible primary balance* (MFPB); therefore, convergence to prudent levels would not be achieved. The maximum initial public debt level at which this problem is not incurred (in 95% of the trajectories) would be the prudent public debt level. A disadvantage of this approach is that, unlike the debt limit level, there are no specific estimates for Chile regarding what the maximum primary balance is.

The probabilistic framework for generating fiscal trajectories will be the same as that proposed in Eyraud et al. (2018), which was developed in Celasun et al. (2006). These authors use a vector autoregression (VAR) to capture the dynamics of non-fiscal variables that affect the debt accumulation equation (domestic and foreign interest rates, exchange rate, and economic growth) and, with a fiscal reaction function, they estimate the response of the primary balance to debt and macroeconomic conditions. After performing 5,000 Monte Carlo simulations combining these elements with shocks to non-fiscal variables (accounting for their correlations) and a fiscal shock incorporating what is not explained by the fiscal reaction function estimation, a range of fiscal trajectories is generated over a 25-year projection horizon. This horizon is longer than the one used in Eyraud et al. (2018), but it provides a broader and more comprehensive view of the evolution of fiscal sustainability in the medium and long term, particularly in light of the demographic situation that will be discussed in depth later.

This paper estimates a prudent level of public debt in Chile with a view to its application in a debt ceiling fiscal rule. The objective of the paper is not to study whether it would be beneficial to establish this rule, but rather to propose a calibration for Chile in general terms (without delving into details such as escape clauses), specifically using the maximum feasible primary balance method of Eyraud et al. (2018), including modifications to capture the particularities of Chile’s situation while remaining aligned with academic literature. This work aims to be a contribution to the Autonomous

Fiscal Council in its analysis of fiscal policy.

The following Section II will review the general literature on the estimation of prudent debt levels, the maximum feasible primary balance, and fiscal reaction functions. Section III details the method to be executed based on Eyraud et al. (2018). In Section IV, a fiscal reaction function is obtained from a panel of countries. Section V estimates the VAR model for the non-fiscal variables. Finally, Section VI presents and discusses the full model estimates and concluding observations.

II. Literature Review

Prudent Debt Levels

A significant portion of the literature on prudent levels of public debt is oriented toward estimating debt limit levels—the level above which a default would occur—and their determinants. While this paper focuses on maximum primary balance levels, both are intimately related.

In Ghosh et al. (2013), the non-linear relationship between the primary balance and the debt-to-GDP ratio is estimated using a fiscal reaction function (which will be explained later). According to the authors, as debt approaches its limit level, interest payments explode, making it more difficult to adjust the balance, leading to a debt spiral until a default occurs—a behavior they describe as fiscal fatigue. Although Ghosh focuses on the implicit debt limit levels in this relationship between primary balances, interest, and debt, under this model there would also exist a maximum capacity to generate primary balances in response to higher debt, i.e., an MFPB.

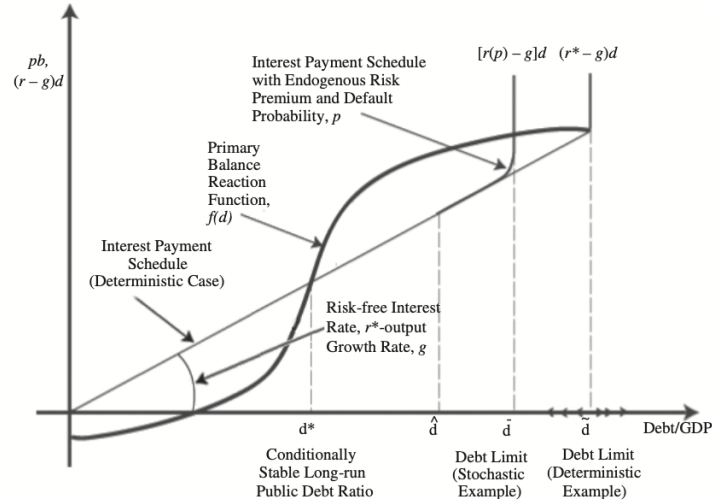


Figure 1: Debt limit under fiscal fatigue. **Source:** Ghosh *et al.* (2013)

Using a general equilibrium model, Mendoza and Oviedo (2009) estimate debt limit levels for middle-income countries, where the government acts as a "tormented insurer" seeking to smooth the con-

sumption of agents but receiving tax revenue subject to exogenous and unexpected shocks. It also incorporates the dollarization of public debt present in many middle-income countries, making borrowing susceptible to exchange rate shocks. Under this model, fiscal sustainability depends not only on the level of debt as a percentage of GDP but also on revenue-raising capacity and its volatility, which would determine the primary balances the government incurs. The debt limit level will be that which ensures the government can meet its obligations under all circumstances, even in the face of large negative shocks.

In Bi (2012), using a general equilibrium model with multiple possible states dependent on domestic variables, rather than estimating a single debt limit, the distribution of the debt limit is estimated as a function of the debt-to-GDP level. This model is consistent with sovereign risk premiums displaying non-linear behavior, wherein above a certain debt-to-GDP level, premiums grow rapidly incorporating the risks.

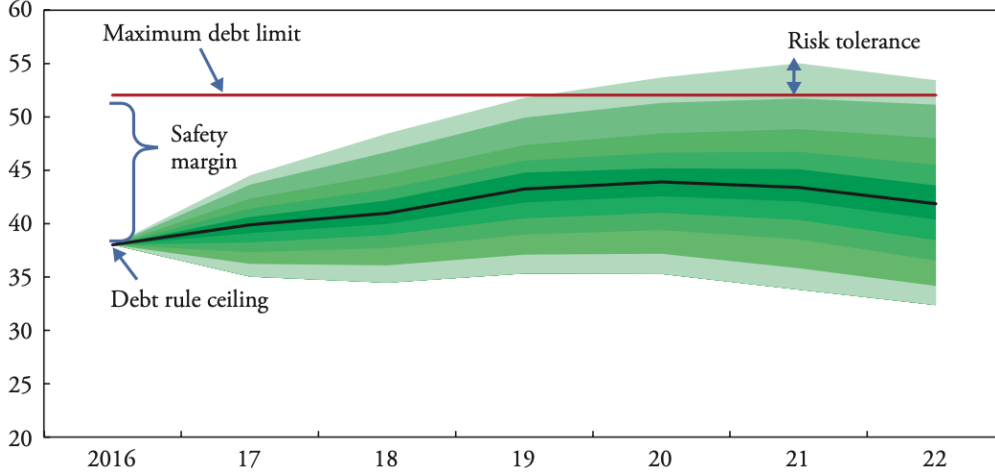
Fournier and Bétin (2018) estimate logistic regressions of default probability based on debt levels, interest rates, and other regressors. Then, combining this result with the no-arbitrage condition (the expected utility of a risk-free bond would be equivalent to one with default risk but higher return), three possible equilibria emerge: low-debt/low-rate scenarios, unstable intermediate equilibria, and equilibria with high debt levels and interest rates featuring elevated risk premiums, from which a debt limit is derived.

Among the specific estimates for Chile, Méndez-Vizcaíno and Moreno-Arias (2021) stand out; replicating Bi (2012), they estimate a debt limit level of 97.1%, which, due to Covid-19 and the significant deterioration in the fiscal situation, fell to 78.9%. Meanwhile, Lozano-Espitia and Julio-Román (2020) and Cruz (2022), using a fiscal reaction function with fiscal fatigue following Ghosh et al. (2013), find levels of 65% and 108%, respectively.

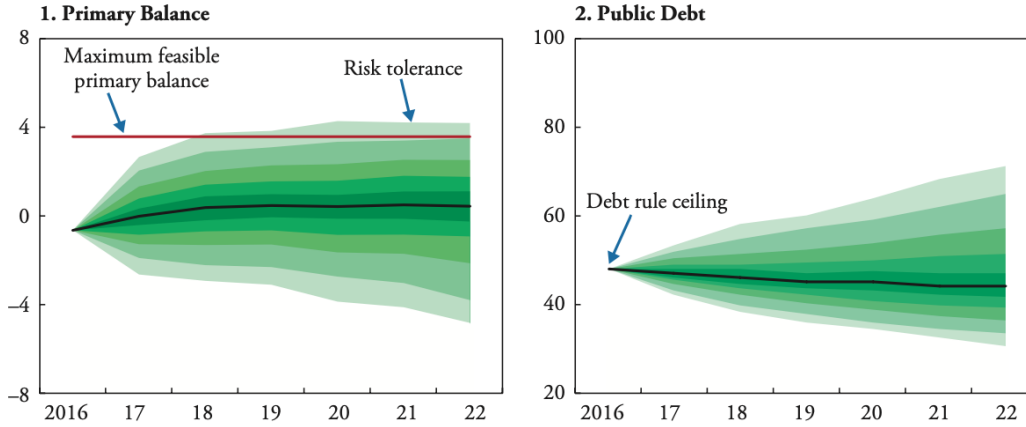
In Eyraud et al. (2018), the IMF first proposes a methodology to estimate a prudent debt level, defining it as the initial debt level that, with a 5% tolerance, ensures the debt limit level is not exceeded across various projected trajectories. In contrast, the second method posited by the authors seeks to provide an approach more tailored to the reality of developed countries, since in these countries it is not clear that a maximum debt limit above which default would occur actually exists. Although Chile is not yet a developed country, this work can still be a novel and relevant contribution without straying from the core discussion. In this method, the prudent level is not derived from the debt limit level; instead, it is defined as the starting point of future trajectories that, despite facing negative shocks, can stabilize without having to undertake excessive fiscal consolidation (exceeding the MFPB) according to historical evidence, with a 5% tolerance.

This approach explicitly incorporates uncertainty into fiscal trajectories through simulations proposed by Celasun et al. (2006). Traditional Debt Sustainability Analyses (DSA) estimate fiscal debt trajectories in the face of standardized deterministic shocks (for example, how it would respond to

a one-standard-deviation drop in growth). In contrast, Celasun et al. (2006) proposes an approach that projects multiple stochastic debt simulations using a VAR for non-fiscal variables, a fiscal reaction function estimated from a panel, and the debt accumulation equation to project fiscal variables forward.



(a) Method 1



(b) Method 2

Figure 2: Eyraud et al. (2018) methods for estimating a prudent debt level

Maximum Feasible Primary Balance

The MFPB would be the maximum consolidation a government could undertake. Theoretically, under a Laffer curve, there would be limits to tax revenue collection. No matter how hard the government tries to increase its revenue, if it is at the peak of the Laffer curve, it will not be able to collect more. Furthermore, there are strong political pressures (whether through interest groups, mobilizations, among others) that prevent public expenditures from being reduced quickly, no matter how much policymakers desire it. Trajectories that exceed these primary balance levels to stabilize

debt would not be attainable; therefore, this limitation must be considered when analyzing public debt sustainability.

The literature on what levels of primary balance might seem excessive has emerged precisely because many countries have high levels of debt and will have to undertake unprecedented fiscal efforts to reach acceptable levels. As Eichengreen and Panizza (2016) point out, the demographic transition and contingent liabilities (explicit or implicit government guarantees) could push debt to even higher levels. Evidence suggests that high primary balances (or "cold showers") are possible but are the exception. The question lies in what that "maximum" level is that would be economically and politically realistic according to historical data.

An initial approximation is performed by the International Monetary Fund (2013) using a sample of 22 countries from 1950 to 2012. The authors take the median of the maximum primary balances for each of the countries as a benchmark. With this, they arrive at a value of 6.5% of GDP for developed countries and 6.25% for developing countries. Along the same lines, in a sample of low-income countries, Baum et al. (2017) define the maximum primary balance as a median across countries of the maximum five-year moving average of country-specific primary balances. This yields values of 4.25% for mineral-exporting countries and 1.25% for non-mineral-exporting countries. Taking historical data for Chile, with data since 1974, the historical maximum was 13.1% in 1998, the 95th percentile sits at 7.6%, Chile's 90th percentile is at 6.2%, and the 75th percentile is at 3.7%. The maximum balance based on historical data is a good alternative, but it does not have to apply under all circumstances, and because it is not cleared of the economic cycle, it may be reflecting an external shock rather than government consolidation efforts. Replicating this procedure with a sample of 120 countries from 1970 to 2020 yields 5.37% for developed countries and 5.18% for developing countries.

Therefore, another alternative is the one followed by Escolano et al. (2014), who, instead of reviewing the maximums, analyzed 91 episodes of fiscal consolidation from 1945 to 2012 (43 in developing countries and 48 in developed countries). They seek to answer: How much, given historical information, is the maximum that a government can adjust in a consolidation process? For this, they define a consolidation episode as those in which the government needs a fiscal adjustment and in which it wants to carry out the adjustment in a discretionary manner.

They establish the first criterion by calculating the difference between the effective primary balance and the one that would be necessary to keep debt stable for the following year. The fact that this difference is greater than 2% of GDP (developed) or 1.8% of GDP (developing) for two consecutive years would reflect that the public sector is far from stabilizing its debt—whether due to unfavorable macroeconomic conditions or poor fiscal management—and that this growth needs to be halted.

They measure the second criterion using the cyclically adjusted primary balance (CAPB) because it reflects the government's intent, rather than changes in economic conditions having provoked a consolidation. The condition is that the change in this indicator is greater than 0.1% of GDP for two

consecutive years. After meeting both conditions, episodes are obtained in which the government must halt its debt growth and where it has made adjustments in its primary balance to achieve it. Subsequently, an episode is considered to continue if its CAPB continues to increase, or if a drop of less than 0.3% of GDP in one year is followed by an increase greater than 0.5% of GDP the next year. The average duration of the studied episodes is 3 years.

Most of the episodes prove successful in halting debt growth, but at a higher level than the initial one. Ordering these episodes by the size of their primary balance at the end of their consolidation yields an MFPB of 2.1% of GDP for emerging economies and 4.5% for advanced economies. This represents an "exceptional" primary balance (the primary balance achieved only by the top 25% of these episodes).

This approach provides a benchmark and characterization of maximum fiscal adjustments, but they are selected based on arbitrary criteria by the authors (incurring a possible selection bias) and may not have been deliberate consolidation episodes. An alternative is to use an "action-based approach" (Devries et al., 2011) which studies each of the episodes using documents from the time to ensure that authorities were seeking consolidation. However, they fall into the problem that all episodes are not treated uniformly, and rigor may be lost when very old episodes are studied.

There is no established methodology to set the maximum level of fiscal consolidation a government can undertake. Nor is it clear that it applies to every country at all times. In Eyraud et al. (2018), it is recommended to use the MFPB estimated in Escolano et al. (2014) to calibrate the model. However, since Chile has achieved highly significant fiscal adjustments compared to other countries (although this also does not imply that it can execute them again in the future), those values would not be adequate (and as will be seen later, they would not fit very well either). In light of this, an MFPB of 5.5% is chosen, which is closer to the 6.25% of the International Monetary Fund (2013), to its replication of 5.18%, and to the fact that Chile historically has higher primary balances than the average of countries.

Fiscal Reaction Function

This regression was introduced by Bohn (1998) to empirically estimate whether the debt of the United States was sustainable. This would occur if increases in the debt-to-GDP ratio due to negative shocks (such as low growth or high interest rates) are met with higher primary balances, controlling for other determinants. Under these circumstances, debt would never exhibit explosive behavior, without the need to make additional assumptions about the interest rate or growth. Subsequent estimates, so as not to be limited by the number of observations, have used panels of countries rather than a single country, though there are also exceptions (Wyplosz, 2005; Burger et al., 2011). Its traditional specification is:

$$pb_{i,t} = \alpha + \rho d_{i,t-1} + \gamma ygap_{i,t} + \beta X_{i,t} + \mu_i + \epsilon_{i,t}$$

Where $pb_{i,t}$ is the primary balance over GDP, μ_i is the country fixed effect, $X_{i,t}$ is a vector of controls, $d_{i,t}$ is the debt-to-GDP level, $ygap_{i,t}$ is the output gap as a percentage of potential output, and $\varepsilon_{i,t}$ is the idiosyncratic component for each country and year.

This equation can additionally include a lag term of the primary balance $pb_{i,t-1}$. This would stem from the nature of the political process, where frictions exist that prevent very abrupt changes, which generates highly persistent primary balance series. Certain authors such as Abiad and Ostry (2005) and Ghosh et al. (2013), instead of including the primary balance lag, use standard errors corrected by Generalized Least Squares (GLS) to allow for error autocorrelation.

In Ghosh et al. (2013), quadratic and cubic terms of the debt lag are incorporated to capture non-linearities that would be empirically observed in highly indebted countries, where above a certain point the capacity to respond with higher primary balances to high debt levels is reduced due to fiscal fatigue. Everaert and Jansen (2018) question the hypothesis of Ghosh et al. (2013), pointing out that the non-linearity actually masks the heterogeneity across different countries. To test this, they estimate the fiscal reaction function using the *Mean Group Estimator* (Pesaran & Smith 1995), allowing each country to have its own ρ_i coefficient.

A problem with this estimation is the possible endogeneity of the output gap with the error term. The determination of the primary balance is affected by the economic cycle—for example, through automatic stabilizers—and therefore this variable is controlled for. However, it is also possible that reverse causality exists, and the determination of expenditures and taxes affects the economic cycle in the other direction. The most frequent solution in the literature (Celasun *et al.* 2006; Everaert & Jansen, 2018; Zeng, 2015) is to use the instrumental variables of the output gap of major trading partners (weighted by their share in each country’s international trade) and the lags of the variable. The problem is that overidentification tests indicate they are weak instruments. Despite this, it is the alternative that will be followed in all estimations.

There is also the possible endogeneity problem generated by the presence of fixed effects in dynamic panel data models. These are included because, in theory, countries that for unobservable time-invariant reasons are capable of generating higher balances on average will tend to have a lower level of debt. Omitting this term would generate a significant bias in ρ and δ . Ghosh et al. (2013) and Celasun et al. (2006)—in one of their specifications—estimate using fixed effects. Applying the *within* transformation, the term $d_{i,t-1} - \bar{d}_i$ would correlate with $\varepsilon_{i,t} - \bar{\varepsilon}_i$, a problem known as Nickell’s bias. However, with longer time horizons this bias would be mitigated¹.

There are different estimation alternatives to eliminate this problem. In Celasun et al. (2006) they propose estimating via *Difference-GMM* (Arellano & Bond 1991) which applies first differences to eliminate the fixed effect and solve the problem of unobserved heterogeneity. It then uses lags of

¹Celasun and Kang (2006) provide a detailed discussion of this bias and the *trade-offs* of using dynamic panel data models instead of fixed effects

the variables in levels as instrumental variables to construct the moment conditions, eliminating the endogeneity that occurs since the term $d_{i,t-1} - d_{i,t-2}$ correlates with $\varepsilon_{i,t} - \varepsilon_{i,t-1}$. Additional lags are also valid instruments, but they can generate overidentification problems, and therefore the number of instruments is limited.

In Baum et al. (2017), as well as in Celasun et al. (2006), it is estimated using *System-GMM* (Blundell & Bond 1998), a method that seeks to improve upon *Difference GMM*, because in the presence of highly persistent endogenous variables (such as $d_{i,t-1}$ and $pb_{i,t-1}$ could be), the instruments can be deficient. This method exploits moment conditions not only in the differenced equation but also in levels, which would solve the problem. However, this comes at the cost of incorporating more instruments, making it more prone to overidentification invalidating the instruments.

It is also fundamental that the selection of control variables aligns with the literature. A very relevant control, particularly in long horizons, is the age dependency ratio, which Ghosh et al. (2013) and Everaert and Jansen (2018) include. This ratio is the ratio between dependents (people under 15 or over 64 years old) and the working-age population (between those ages). In Everaert and Jansen (2018), they use the dependency ratio only for people over 64 years old. A smaller working-age population (higher dependency ratio) would imply relatively fewer workers in the economy, lower production, and therefore lower revenue. But on the other hand, worse *future* demographics should also stimulate *forward-looking* governments to improve their balances today to cushion future costs, and for the same reason, the age dependency ratio in 20 years (or its projection) is included as a control.

However, both authors estimate negative effects for the coefficient associated with the future population dependency ratio (whether total or only older adults), but fail to find this effect. This can be explained because it is capturing the *dynamic* effect that the transition toward that new demographic point will have. Subject to the same dependency ratio today, it is different whether in the future there is a transition to a low or a high dependency ratio. The first case would mean that the country is in a demographic position of growth, with many individuals under 15 years old and where a *demographic dividend* exists. This is a boost in economic productivity that occurs when there is a growing number of people in the workforce relative to the number of dependents². In this demographic position, it makes sense to make investments because markets will expand moving forward. The second case would reflect the exact opposite: an aging population with a shrinking workforce, which translates into a *demographic burden* due to a lower economic momentum and higher health and social security expenditures. For this reason, it makes sense to add a second future term in addition to the contemporary dependency ratio, since the composition of the dependent population can be different, and this can affect the demographic transition.

Another relevant control is the terms of trade. Abiad and Ostry (2005) and Baum et al. (2017)

²Definition by the [United Nations Population Fund](#)

include it as the gap relative to the trend terms of trade. Eyraud et al. (2018) also recommend the inclusion of this term in the estimation of the fiscal reaction function for developing countries, given the high dependence their fiscal revenues may have on this variable. Both authors use dummy variables indicating whether they are commodity-exporting countries or not, to separate both effects³. Terms of trade cycles do not necessarily have to move together with general economic cycles, and in countries highly dependent on commodity exports, this inclusion would be justified. Changes in the prices of these commodities—in the case of Chile, especially copper—undoubtedly affect general macroeconomic conditions and fiscal revenues, for example through specific taxes on natural resources or state-owned enterprises.

III. Methodology

The methodology developed to generate probabilistic simulations for debt and the primary balance in terms of GDP is detailed below. The work of Celasun et al. (2006) is used as a guide, but innovations are included for the case of Chile. Among these, incorporating the lag of the primary balance in the projection of the primary balance through the fiscal reaction function, which is something frequently added when estimating this function but that the original model does not consider. As the fiscal reaction function is used in this work as an instrument for subsequent simulations, the inclusion of this term turns out to be very relevant to smooth the trajectories⁴. Second, the inclusion of the terms of trade to capture the importance of commodity price movements in public finances. Finally, the incorporation of the population dependency ratio, since given the long time horizon used, demographic trends of population aging must be considered.

First, a *reduced-form VAR* is estimated for the non-fiscal variables as follows:

$$Y_t = \gamma_0 + \sum_{p=1}^P \gamma_p Y_{t-p} + \xi_t, \text{ where } Y_t = (r_t^d, r_t^f, reer_t, g_t, tot_t)$$

Where r_t^d is the short-term domestic real interest rate, r_t^f is the US real interest rate, $reer_t$ is the logarithm of the real effective exchange rate, g_t is real economic growth, and tot_t is the terms of trade index. Section IV provides more details on this estimation. This VAR is estimated on a quarterly basis to increase its robustness, but the projections must then be annualized to connect them with the fiscal variables, which are in annual terms. The annualization of interest rates and growth is done through a rate conversion⁵, while the average of the last four quarters is used for the exchange rate and the terms of trade index. The projections of these variables for T periods ahead are carried out using Monte Carlo simulations, which, knowing the estimated parameters, incorporate into each

³In this work, the reaction function does not include the term multiplied by the *dummy* for non-exporting countries as it is not of interest, but it does not alter the results significantly

⁴Simulations without this term respond much more aggressively to higher debt, which can be somewhat unrealistic

⁵ $y_t^{annual} = (1 + y_t)(1 + y_{t-1})(1 + y_{t-2})(1 + y_{t-3}) - 1$

of the K simulations a series of shocks $\xi_{t,k} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_y^2)$ with variance σ_y^2 coming from the estimated variance-covariance matrix. In this way, the correlation between shocks to different variables is also taken into account. Then, K distinct trajectories of T periods are generated for these variables, which would be consistent with historical evidence.

Second, a *Fiscal Reaction Function* is estimated using a panel of I countries, capturing the response of the primary balance to the previous debt level and other variables. The chosen specification is the following:

$$pb_{i,t} = \alpha + \delta pb_{i,t-1} + \rho d_{i,t-1} + \gamma ygap_{i,t} + \tau totgap_{i,t} D_{exp} + \beta_{ADR} adr_{i,t} + \beta_{FADR} adrold_{i,t+20} + \mu_i + \varepsilon_{i,t}$$

Where $pb_{i,t}$ is the primary balance over GDP, $d_{i,t}$ is the debt level over GDP, $ygap_{i,t}$ is the output gap as a percentage of potential output, $totgap_{i,t}$ is the terms of trade gap relative to its trend level, D_{exp} is a dummy variable indicating whether it is a commodity-exporting country or not, $adr_{i,t}$ is the population dependency ratio, $adrold_{i,t+20}$ is the population dependency ratio for those over 64 years old in 20 years, μ_i is the country fixed effect, and $\varepsilon_{i,t}$ is the idiosyncratic error term for each country and year. Details of this estimation are explored further in Section IV.

Subsequently, for each of the K simulations, starting from a known initial debt level d_0 and primary balance pb_0 , and a fixed percentage of dollar-denominated debt θ , a debt level is projected for the following period. This is done using the VAR projections for the non-fiscal variables⁶ in the fiscal reaction function and in the *public debt accumulation equation* (the government's intertemporal budget constraint):

$$d_{t,k} = \frac{[\theta(1 + r_{t,k}^f)(1 + \Delta reer_{t,k}) + (1 - \theta)(1 + r_{t,k}^d)]}{(1 + g_{t,k})} d_{t-1,k} - pb_{t,k}$$

The projection algorithm follows a recursive structure. The debt level resulting from the debt accumulation equation is used in the next period to project the primary balance, and then this primary balance is used to obtain the debt level, and so on. Specifically, for each of the K simulations, for each of the T periods, there will be estimated values of the non-fiscal variables (which will be different for each simulation). Then, using the parameters from the estimation of the fiscal reaction function, the primary balance for that period and simulation is obtained. The previous values of the primary balance $pb_{t-1,k}$ and debt $d_{t,k}$ would be known. The output gap $ygap_{t,k}$ is calculated as the difference between growth and the steady-state growth implicit in the VAR ($ygap_{t,k} = g_{t,k} - g^{ss}$). The terms of trade gap $totgap_{t,k}$ is calculated as the ratio between the terms of trade index and the one implicit in the steady state minus one ($totgap_{t,k} = tot_{t,k}/tot^{ss} - 1$). The population dependency ratios adr_t and $adrold_{t+20}$ will be the same for all K simulations and come from United Nations projections. The term μ_{CL} would be fixed for all simulations and periods and would represent other

⁶The differential of the logarithm of the real effective exchange rate relative to the previous period is used

unobserved factors that differentiate Chile's fiscal policy from that of the panel of countries. Finally, shocks $\varepsilon_{t,k}^f \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_f^2)$ are incorporated, with variance σ_f^2 equal to the variance of the residuals from the estimation of the fiscal reaction function for Chile's values; in this way, the historical variation that was not captured in the estimation of the fiscal reaction function is incorporated.

The method developed by Eyraud et al. (2018) combines this construction of trajectories with the notion that a maximum feasible primary balance (MFPB) would exist; therefore, trajectories in which this level is exceeded would not be realistic (even if generated using a correctly specified VAR estimation and fiscal reaction function) and should be considered exceptional. Therefore, the initial debt level d_0 must be iteratively altered until the condition that only 5% of the trajectories surpass the MFPB is met. Through this procedure, a prudent debt level d_0^P is reached.

It is crucial to point out the deficiencies and criticisms of this model. First, it is not consistent with reality that non-fiscal variables like interest rates are not affected by fiscal variables, since as the public debt level increases, it is evident that international agents will demand a higher interest rate that incorporates the probability of that country defaulting. Annex 6 presents a way to address this problem.

Second, another criticism comes from the trajectory estimation method. Monte Carlo simulations must assume a probability distribution, in this case Gaussian, which would limit the trajectories of fiscal and non-fiscal variables to being symmetric. This does not necessarily happen in reality, where trajectories for different variables can be much more skewed in one direction than the other. In contrast, using bootstrapping or asymptotic distributions would not fall into this problem because no prior distribution has to be assumed.

On the other hand, there would be more generalized criticisms of reduced-form estimations. Following the Lucas critique, parameters estimated today based on historical data may cease to be realistic in the future. Instead of estimating in this way, the correct approach would be to calibrate, using structural models, the maximum consolidation adjustment that the government could make, and thereby estimate prudent levels. This way, expected structural changes in the economy would be incorporated. However, these models require more assumptions that could be unrealistic and do not guarantee robust results either.

Finally, there is the criticism that the model assumes that the potential output growth from which the output gap is calculated will always be the same (the steady-state growth implicit in the VAR), and the same applies to the terms of trade gap: it assumes that the gap is always calculated in comparison to the same trend level (steady-state terms of trade). This ensures that in the steady state, these terms do not significantly alter the fiscal variables⁷. Nor is it sustainable to assume that the percentage of dollar-denominated debt is fixed throughout the entire projection horizon. As a country becomes less reliable to international investors, they will demand dollar-denominated debt

⁷See Annex 7

to eliminate risks. These are inherent limitations of the model that would further favor a general equilibrium approach.

IV. Fiscal Reaction Function Estimation

For the estimation of the fiscal reaction function, five alternative specifications are proposed which, in addition to the econometric differences discussed in the literature review, allow incorporating information specific to the Chilean situation through a fixed effect without losing the robustness of a panel estimation. All estimations instrument the output gap with two of its lags and with the output gap of its commercial partners, weighted by their relevance in each country's international trade.

Regarding the data, with the exception of the dependency ratio data, which comes from the United Nations, the data used are from the International Monetary Fund⁸. The selected sample includes 105 countries from 1985 to 2020; years prior to this were not considered as the volatility of the series increases considerably and the data could be less precise. The output gap was estimated using a Hodrick-Prescott filter when it was not available in the data. The terms of trade gap was estimated using a Hodrick-Prescott filter. Annex 4 includes robustness exercises regarding the samples of years and countries, to limit them to periods and countries that are representative of Chile's situation.

Specification (1) is estimated using *Difference-GMM* and (2) using *System-GMM*. To avoid instrument proliferation problems, both the difference and level equations are limited to 2 lags. Specifications (3) and (4) are estimated using fixed effects, and since the panels are long (averaging 27.6 years), the effect of the Nickell bias should be limited. Specification (4) follows Ghosh et al. (2013) and seeks to capture fiscal fatigue by including quadratic and cubic coefficients for the debt lag. Finally, specification (5), like Everaert and Jansen (2018), estimates heterogeneous coefficients by country for $d_{i,t-1}$ using a *Mean Group Estimator*; this way, the correlation with the fixed effect would be eliminated since each country would have its own coefficient.

Ghosh et al. (2013) include the dependency ratio $adr_{i,t}$ and its value in twenty years $adr_{i,t+20}$, while Everaert and Jansen (2018) use the dependency ratio of those over 64 years old $adrol_{i,t}$ and its value in twenty years $adrol_{i,t+20}$. Both specifications were tested, but the inclusion of the same variable measured at two different points in time generates multicollinearity. This problem is caused by an unobservable time-invariant common factor, in this case, the variables that cause the demographic transition. The presence of this multicollinearity will cause that as the correlation between the variables increases, their estimated coefficients will misleadingly tend toward infinite magnitudes in opposite directions. Given this, a combination of both approaches is used: to capture the effect of having a larger or smaller working-age population today, $adr_{i,t}$ is incorporated, while to add the

⁸Annex 1 details the sources: World Economic Outlook and Public Finances in Modern History

demographic burden of having an aging population (in the case of Chile), $adrol d_{i,t+20}$ is used. Annex 3 details different specifications to address this problem and successfully incorporate demographic variables into the fiscal reaction function, a topic rarely explored in the literature.

The estimation presented in Table 1 yields similar results across the different specifications. The most relevant finding is that the coefficient for the public debt lag is positive and significant in all proposed specifications, which is in line with the literature, indicating that on average countries respond by combating debt increases with higher primary balances. The magnitude of this coefficient is usually around 0.05 - 0.03 ((2) and (3) would be underestimating). It is also worth noting that in (5), the heterogeneous coefficient estimated for Chile is larger than that of the rest of the sample, indicating that Chile responds more strongly to higher debt than the rest of the panel.

The high and significant coefficient of the primary balance (except in (5)) is also in line with the literature. While the output gap term is usually between 0.1 - 0.3, with only estimates (1) and (2) being consistent with this. Regarding the terms of trade gap, Abiad and Ostry (2005) estimate an associated coefficient of 0.1 but also include the price of oil separately. Baum et al. (2017) find similar coefficients but in estimations for low-income countries. The signs and significance found are consistent with this, but the magnitude would be excessive.

The negative effect of the dependency ratio and the elderly dependency ratio (except in (3) and (4)) is in line with the findings of Ghosh et al. (2013) and Everaert and Jansen (2018), indicating that a larger dependent population today generates lower primary balances and that if the population is aging, primary balances will be lower than with a growing working-age population. As will be seen later, the problem lies in ensuring that this demographic pressure does not become excessive.

It is important to highlight the fixed effect, which for all estimations is positive and averages 1.5%, reflecting that Chile tends to be among the countries whose unobserved factors push them to execute higher primary balances than those in the sample, which would be in line with establishing a relatively high maximum feasible primary balance (MFPB). On the other hand, the variance of the residuals for Chile (the variance of the fiscal shock) is greater than the mean squared error, reflecting that Chile fits the model relatively worse compared to other countries in the panel.

For models (1) and (2), it is necessary to run autocorrelation tests, which show first-order autocorrelation (which happens since first differences are applied) but reject second-order autocorrelation (which would be a problem because it would violate the sequential exogeneity assumption). Sargan-Hansen tests must also be applied, which would affirm that the instruments used are valid and there are no overidentification problems.

For the fixed effects models (3) and (4), the Sargan-Hansen tests reject the validity of the instruments (two lags of the gap and the weighted output gap of trading partners), indicating that the instruments would not be valid, which is in line with what is found in the literature. This result is compatible with the previous one since all instruments must be tested together, regardless of which variable

Table 1: Fiscal reaction function estimates

	(1) Diff-GMM	(2) Sys-GMM	(3) FE	(4) FE2	(5) MG
$pb_{i,t-1}$	0.4887*** (0.0049)	0.6297*** (0.0043)	0.6120*** (0.0151)	0.6108*** (0.0152)	0.5655 (1.2643)
$d_{i,t-1}$	0.0277*** (0.0004)	0.0211*** (0.0004)	0.0151*** (0.0018)	0.0389*** (0.0057)	0.0370*** (0.0043)
$ygap_{i,t}$	0.1839*** (0.0025)	0.1656*** (0.0048)	0.0161 (0.0171)	0.0209 (0.0173)	0.0894 (0.3494)
$totgap_{i,t} D_{exp}$	0.5861*** (0.0091)	0.3584*** (0.0334)	0.4991*** (0.0491)	0.4957*** (0.0489)	0.5600 (2.8707)
$adr_{i,t}$	-0.0263*** (0.0043)	-0.0651*** (0.0030)	0.0090 (0.0077)	0.0102 (0.0077)	-0.0010 (0.3924)
$adrol d_{i,t+20}$	-0.1281*** (0.0046)	-0.0847*** (0.0028)	-0.0868*** (0.0084)	-0.0876*** (0.0084)	-0.1298 (0.6990)
$d_{i,t-1}^2$				-0.0002*** (0.0000)	
$d_{i,t-1}^3$				0.0000*** (0.0000)	
Constant	2.4906*** (0.3017)	4.3574*** (0.2294)	0.1582 (0.5187)	-0.5875 (0.5447)	0.5685* (0.2321)
Observations	2844	2947	2901	2901	2901
Countries	105	105	105	105	105
Mean squared error	3.19	3.68	4.45	4.41	3.92
F statistic	10561	10519	379	289	22
Instruments	143	207			
AR(1) p-value	0.00	0.00			
AR(2) p-value	0.50	0.31			
Sargan–Hansen p-value	0.98	1.00	0.00	0.00	
F p-value $d_{i,t-1}^2, d_{i,t-1}^3$				0.00	
Residual variance Chile	3.7986	4.2054	4.4479	4.7415	4.5433
FE Chile	1.7471	0.8218	1.4410	1.7502	1.8790
$d_{i,t-1}$ Chile					0.0448

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors in parentheses

(1), (2) and (3) use robust standard errors; (4) clusters by country

they seek to instrument. The F-test for the joint significance of $d_{i,t-1}^2$ and $d_{i,t-1}^3$ rejects that debt has a linear behavior.

Despite having the advantage of incorporating more Chile-specific information through the debt lag coefficient, model (5) turns out to be the weakest, with a lower joint significance than the rest of the models and only the heterogeneous coefficient and the constant being significant. For model (5), there were no overidentification tests available, but the output gap was instrumented.

Specification (1) is the one that best fits Chile's situation. Its coefficients are all significant and in line with what is expected by the literature, and therefore it will be the baseline specification to follow in this model.

V. VAR Estimation for Non-Fiscal Variables

The parameters selected to estimate the reduced-form VAR indicated in Section III are the same as Celasun et al. (2006), with the exception of the incorporation of the net commodity export price index, which reflects the terms of trade. The inclusion of this term in the VAR is justified since in Chile, the terms of trade affect the other variables of interest (exchange rate, interest rate, growth) and this would strengthen their projections. Projecting this variable separately would also require having projections for copper and oil prices, which may cease to be relevant for Chile in the future and which are also non-stationary variables.

The short-term domestic interest rate is slightly different from the one used by the authors. For this, the interest rate of the 90-day discounted promissory notes of the Central Bank of Chile (PDBC) in pesos in the primary market (the SOMA system) was used as the short-term interest rate. These rates are adjusted for inflation to obtain real interest rates; this adjustment and the data sources are detailed in Annex 2.

To provide greater realism, restrictions are incorporated into the VAR coefficients, limiting the effects of domestic variables (domestic real interest rate, real effective exchange rate, real growth) on international variables (US real interest rate, terms of trade index). This transitions the estimation from OLS to Feasible Generalized Least Squares (FGLS).

It is not necessary to use a structural VAR model for the projection of variables, nor would it alter the projections. Solving the identification problem and transitioning to a structural VAR would only be justified to seek to estimate causal relationships between shocks of each variable and the other variables, such as with impulse-response functions.

Given the exceptional situation caused by the Covid-19 pandemic, the last two available years were omitted from the projections since they are outliers and significantly alter the estimation. Under information criteria, the VAR(1) turns out to be the best specification; its residuals are white noise

and it meets the stability condition (tested using the *varsoc* and *varstable* commands in Stata, detailed in Annex 2). The fit of the VAR model to the non-fiscal variables is shown below:

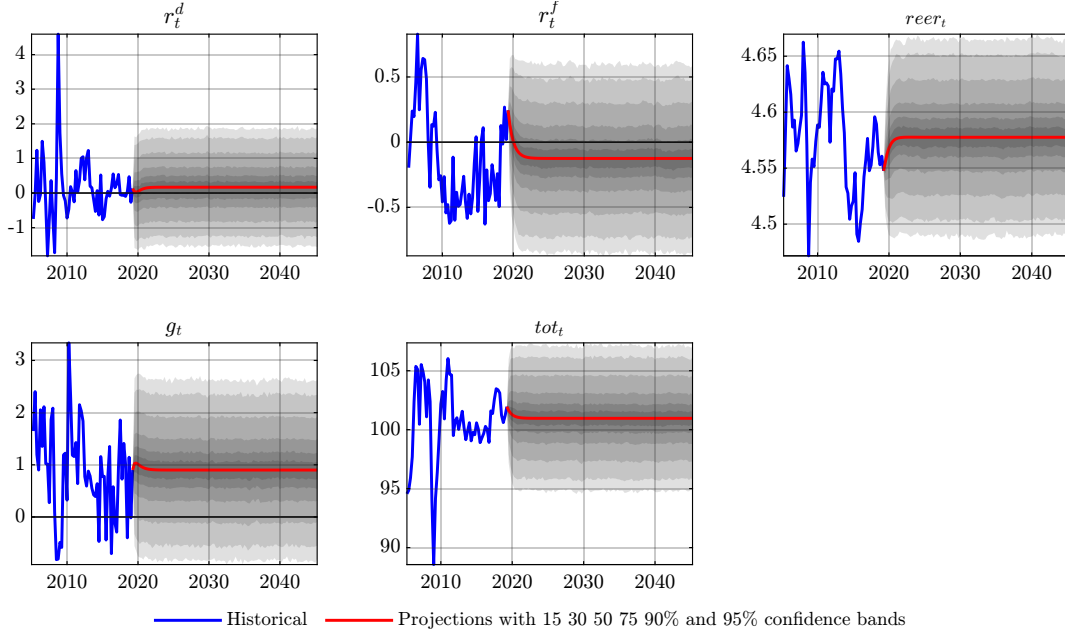


Figure 3: VAR projections for non-fiscal variables

VI. Estimation of a Prudent Public Debt Level

Below are estimations of prudent public debt levels in Chile. They were carried out with 5,000 Monte Carlo simulations using the proposed methodology. The share of dollar-denominated debt was fixed at its 2022 value of 38.0%⁹. For the dependency ratio trajectory and its future value, the United Nations baseline projection up to 2050 was used, and then it stabilizes at that level. For the purposes of this work, the prudent debt level is defined as the highest debt level compatible with a primary balance trajectory that in 95% of the cases does not exceed a maximum feasible limit of 5.5% of GDP in 25 years. This MFPB was established based on the maximum primary balances literature (replicating International Monetary Fund (2013) yielded a MFPB of 5.18%) and Chile's own data, which indicated that Chile would have a relatively higher MFPB.

Figure 4 shows the construction of the debt and primary balance trajectories, with confidence bands for the simulations generated from the fiscal reaction function (3) (as an example). Additionally, two of the 5,000 generated simulations are added to show their evolution. As seen in the blue trajectory, it is possible for large primary balances to occur in the future; what is relevant is to be certain that this would be an extreme case that does not occur in the majority of the simulations. The confidence

⁹Dipres (2022). 26.3% is denominated in USD and 11.7% in EUR; for simplicity, all of it is assumed to be in USD

bands allow exhibiting this uncertainty in the simulations.

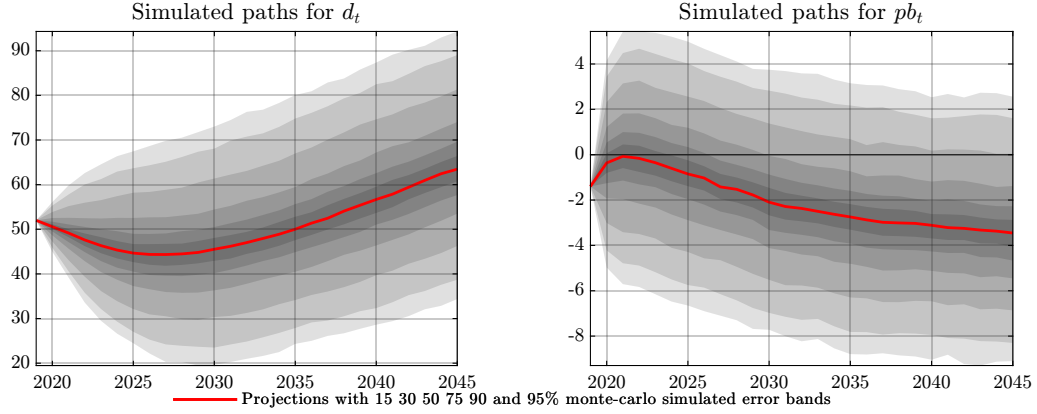


Figure 4: Projections from specification (3) starting at the 38.2% baseline scenario

Figure 5 shows debt trajectories (the central scenario and bands reflecting 95% confidence) using specification (1). In red, the trajectories starting from the current debt level with debt at 38.2% are shown, and in blue, the projected trajectories if the debt were at the "prudent" level (the debt ceiling) of 51%. As this initial debt level of 51% is exceeded, the primary balance response will be greater (connecting through the budget constraint and the fiscal reaction function) and eventually a significant percentage of the primary balance trajectories would surpass the MFPB, which should not be possible given historical experience.

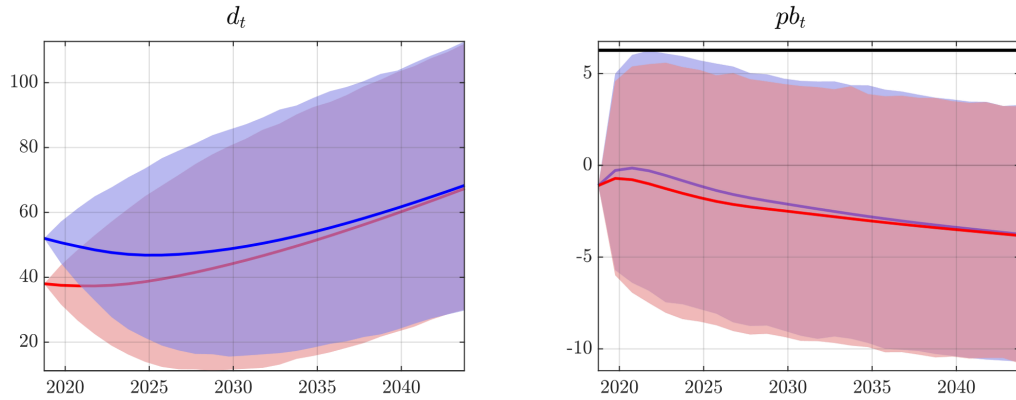


Figure 5: Projections from the 38.2% baseline scenario (red), prudent level of 51% (blue), and MFPB of 5.5% (black)

Table 2 shows the replication of this exercise for the different fiscal reaction function specifications; it is also shown for a MFPB of 5.75% and 6.00%. It is observed that small changes in the MFPB generate significant changes in the prudent level. Since the MFPB for Chile is not known, this reflects a weakness of the model compared to other approaches focused on the debt limit level (which is known). The prudent debt levels obtained of 51% in (1), 47% in (2), and 45% in (3) are reasonable, while those resulting from specification (4) and (5) are relatively low (considering that the debt

today stands at 38.2%).

Table 2: Prudent initial debt level for different MFPB values

	(1)	(2)	(3)	(4)	(5)
MFPB	Diff-GMM	Sys-GMM	FE	FE2	MG
5.50%	51%	47%	45%	32%	42%
5.75%	58%	53%	54%	37%	45%
6.00%	62%	59%	62%	42%	55%

The differences in the prudent levels obtained are primarily due to different ρ ¹⁰, since if the response to higher debt levels is stronger, it will be more likely that the MFPB is exceeded at lower debt levels. Even if the fiscal reaction function coefficients are relatively similar, small changes can generate large differences; for example, if the debt level is 50%, a 0.02 increase in ρ generates an additional primary balance response of 1 percentage point of GDP each year.

Second, the larger fiscal shock $\varepsilon_{t,k}^f$ resulting from these estimations is also relevant, which would widen the 95% confidence bands and make it more likely to exceed the MFPB at lower debt levels. It is correct to use different fiscal shocks for each specification because they represent how well the specification fit Chile's historical data. For instance, perhaps the higher residual variance of specification (4) comes from the fact that in historical situations of high debt, the balance response was not in line with what was expected according to the coefficients of $d_{i,t-1}^2$ and $d_{i,t-1}^3$, and the higher shock variance would allow capturing those differences.

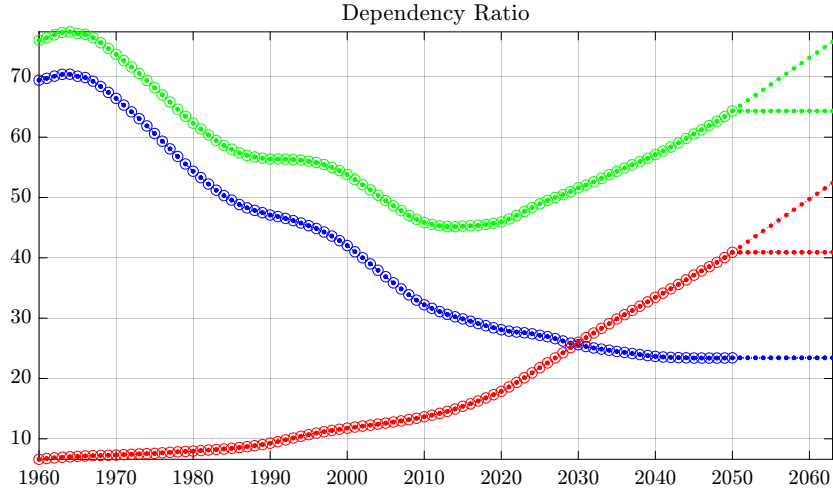


Figure 6: Dependency ratio: under-15 (blue), over-64 (red), and total (green)

It is important to point out that both trajectories in Figure 5 converge towards the end of the graph. This reflects that as the primary balance response due to higher debt stabilizes, the trajectories

¹⁰In (4), the coefficients of $d_{i,t-1}^2$ and $d_{i,t-1}^3$ also play a role

tend to the same convergence level regardless of the initial debt level. Then the question will be what the steady-state debt level is and whether it will be above the debt ceiling. To project forward, assumptions must be made about the evolution of demographic variables, since these are only available until 2050. Figure 6 shows various possible trajectories that these variables can follow. The upper trajectory assumes that the dependency ratio for those over 64 years old follows a linear trend, the central one assumes that it stabilizes at 50% by the year 2080, and the lower one takes the last value of 2050 for the entire projection.

These dependency ratio trajectories strongly affect the evolution of fiscal variables. Figure 7 shows the evolution of public debt using the reaction function specification (5) in the three different scenarios. In red, the evolution of debt is shown if the dependency ratio follows the linear trend. In green is the trajectory if the dependency ratio stabilizes at 50% by 2080. In blue, the trajectory of public debt is observed if it remains at the 2050 level. Debt tends to increase the worse the demographic situation gets, consistent with what is observed in countries with advanced aging situations like Italy or Japan.

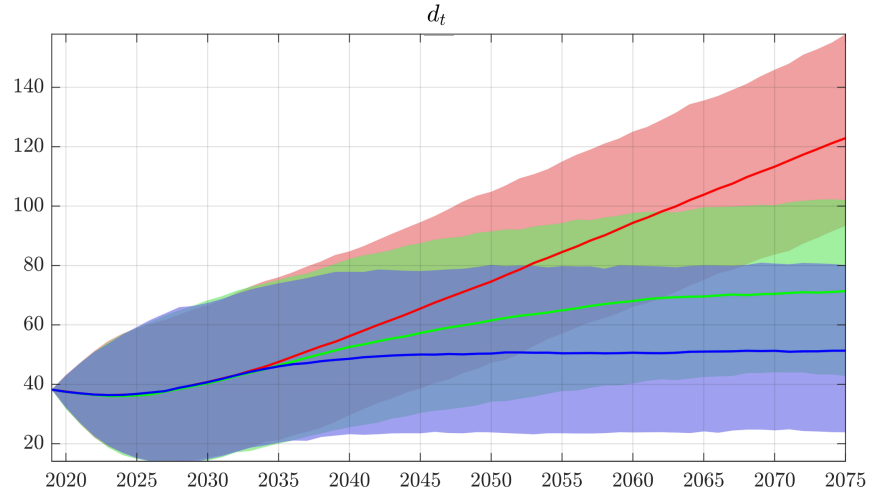


Figure 7: Debt trajectories if the dependency ratio follows a linear trend (red), stabilizes in 2080 (green), or stays at its 2050 level (blue)

Table 3 exhibits the steady-state convergence levels for all fiscal reaction function specifications. The procedure for calculating these levels is detailed in Annex 7.

Table 3: Steady-state convergence levels of public debt to GDP

	(1)	(2)	(3)	(4)	(5)
Dependency ratio	Diff-GMM	Sys-GMM	FE	FE2	MG
Linear projection*	152.23%	193.35%	123.08%	134.49%	108.09%
Stabilization at 50%	91.97%	115.14%	74.57%	65.07%	69.61%
2050 levels	60.51%	74.31%	49.24%	43.02%	49.53%

*For the linear projection, the 2080 level is shown, since it does not converge

As observed, the resulting debt trajectories do not necessarily converge; rather, convergence depends on whether the dependency ratio eventually stabilizes and on the level at which this occurs. This reflects a major weakness of the proposed MFPB methodology: even if the prudent debt level is successfully estimated in the medium term (the authors propose using 6 years of projection) ensuring that the MFPB is not exceeded, this does not ensure that such debt trajectories are sustainable in the long run or that convergence will eventually occur. A debt ceiling seeks to eliminate the few cases in which debt exceeds certain undesirable levels (such as those where a default occurs), but if debt in all its trajectories tends to exceed this level, then a problem would exist. If this occurs, after a few years the pressure to not respect a 51% debt ceiling would be enormous, since most trajectories exceed this level and it would not be clear whether that is sustainable or not.

However, this does not invalidate the conclusions of the proposed model, which was designed as a simple medium-term model that does not seek to capture all the complexities of fiscal policy, but rather to serve as an additional tool. Although the model shows that long-term convergence depends very heavily on the stabilization of demographic variables, these parameters are difficult to estimate and it is likely that the model loses its predictive capacity over very long time horizons for which it was not designed. Important risks may be being underestimated—for example, costs associated with climate change mitigation and adaptation—or overestimated—such as the demographic transition, which, due to a *second* demographic dividend (a population concentrated in older ages has incentives to accumulate assets, which can be invested and generate more wealth for society; these results are not automatic but depend on the implementation of effective policies that foster capital accumulation), may not be crucial for fiscal trajectories if appropriate policies are taken.

Assessing prudent debt levels for a particular country requires a deep understanding of the fiscal context to properly identify the upside or downside risks of establishing a debt ceiling. As observed, the model's projections indicate that small changes in the calibration of the MFPB generate significant differences in the obtained prudent levels. Given this, finding a way to estimate this parameter is a priority.

The situation of growing public debt and interest payments in Chile must be addressed. To improve the country's fiscal framework, this document suggests establishing a debt ceiling at a level of 51% of GDP, consistent with a 95% probability of not exceeding a maximum primary balance of 5.5% over a 25-year period. The methodology used is framed within the literature on fiscal rules and fiscal sustainability, and seeks that, given the wide spectrum of possible scenarios, this rule sets a debt ceiling *compatible with realistic future fiscal consolidations* without being extremely rigid and maintaining simplicity to communicate to citizens. As has been discussed, the model is very fragile to changes in assumptions and it is encouraged to use it as an additional input, combining it with other analyses and in-depth research of the country's situation.

Despite the results obtained, this work considers that the establishment of a borrowing rule is merely one more measure to improve the fiscal situation; it is its combination with other measures

within a solid institutional framework that allows achieving fiscal sustainability. It is concluded that the establishment of a dual rule to guide the actions of fiscal policy can indeed contribute to macroeconomic stability in Chile and strengthen the fiscal framework.

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Annexes

Annex 1: Sources and definitions for the reaction function

Annual fiscal data for the panel: World Economic Outlook (WEO), April 2022 (available at <https://www.imf.org/en/Publications/WEO/weo-database/2022/April>) and Mauro et al. (2013), Public Finances in Modern History (PFMH), available at <https://www.imf.org/external/np/fad/histdb/>. Variables used: output gap (percent of potential GDP), real GDP growth, general government primary balance (percent of GDP), and general government gross debt (percent of GDP).

Age dependency ratio: World Bank WDI, based on UN Population Estimates and Projections (available at <https://databank.worldbank.org/source/population-estimates-and-projections>). The future dependency ratio was defined, as in Ghosh et al. (2013), as the ratio 20 years ahead.

Export and import data: Direction of Trade Statistics, IMF. Used to construct the instrument for $ygap$, with the output gap of trading partners weighted by their relevance in each country's international trade:

$$ygap_{i,t}^Z = \frac{\sum_{j \neq i}^N s_{j,t} ygap_{j,t}}{\sum_{j \neq i}^N s_{j,t}}$$

The following 15 countries were used: United States, China, Germany, Japan, United Kingdom, France, Netherlands, Italy, India, Russia, Belgium, Hong Kong, Spain, Brazil, and South Africa.

Annex 2: Sources, definitions, and FGLS estimation of the VAR model

Quarterly real growth: Central Bank of Chile

Quarterly real effective exchange rate: International Financial Statistics, IMF

US GDP deflator: International Financial Statistics, IMF

International interest rate: 3-month US Treasury bill rate. Federal Reserve Economic Data, St. Louis Fed

Domestic interest rate: interest rate on 90-day peso-denominated discountable promissory notes of the Central Bank of Chile (PDBC) in the SOMA system. To convert to real interest rates, the formula proposed by Celasun et al. (2006) was used:

$$r_t = \frac{(1 + \text{IntRate}_t/100)^{0.25}}{1 + \text{CpiInflation}_{t+1}} - 1$$

The estimation satisfies the stability conditions (eigenvalue moduli below one). The autocorrelation test on the residuals does not reject the null hypothesis of no autocorrelation. According to the HQ and SB information criteria, the VAR(1) would be optimal, while only the AIC indicates that a VAR(2) would be optimal.

VAR coefficients

	Domestic interest rate	Foreign interest rate	Log REER	Growth	ToT index
Domestic interest rate	0.2593	0.0000	0.0095	-0.1053	0.0000
Foreign interest rate	-0.0443	0.7058	-0.0007	0.3200	-0.0765
Log REER	3.9352	0.0000	0.6616	0.3976	0.0000
Growth	-0.1690	0.0000	0.0162	0.2860	0.0000
ToT index	0.0095	0.0000	-0.0013	0.0138	0.7077
Constant	-18.7035	-0.0367	1.6649	-2.5094	29.5060

Residual variance-covariance matrix

	Domestic interest rate	Foreign interest rate	Log REER	Growth	ToT index
Domestic interest rate	0.6586	0.0386	-0.0036	-0.0216	-0.8808
Foreign interest rate	0.0386	0.0682	-0.0003	-0.0454	0.0272
Log REER	-0.0036	-0.0003	0.0006	0.0076	0.0145
Growth	-0.0216	-0.0454	0.0076	0.6835	0.1489
ToT index	-0.8808	0.0272	0.0145	0.1489	4.9433

Steady state (quarterly)

Domestic interest rate	Foreign interest rate	Log REER	Growth	ToT index
0.1681	-0.1246	4.5776	0.8994	100.9618

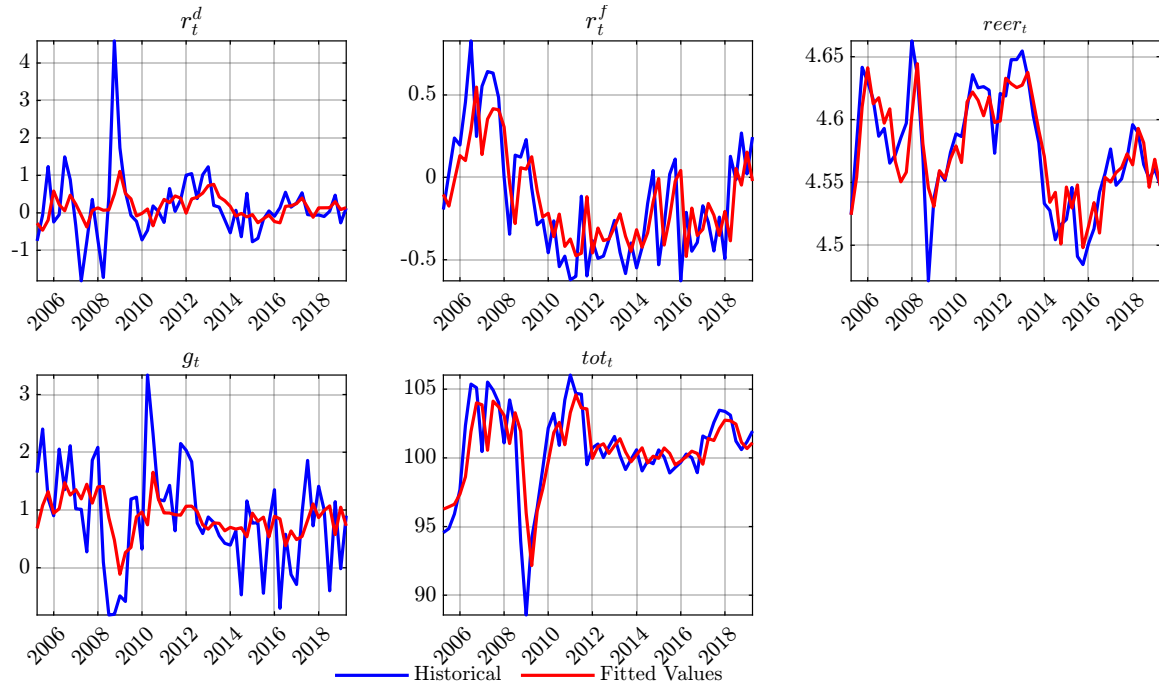


Figure A2.1: Fitted and actual values

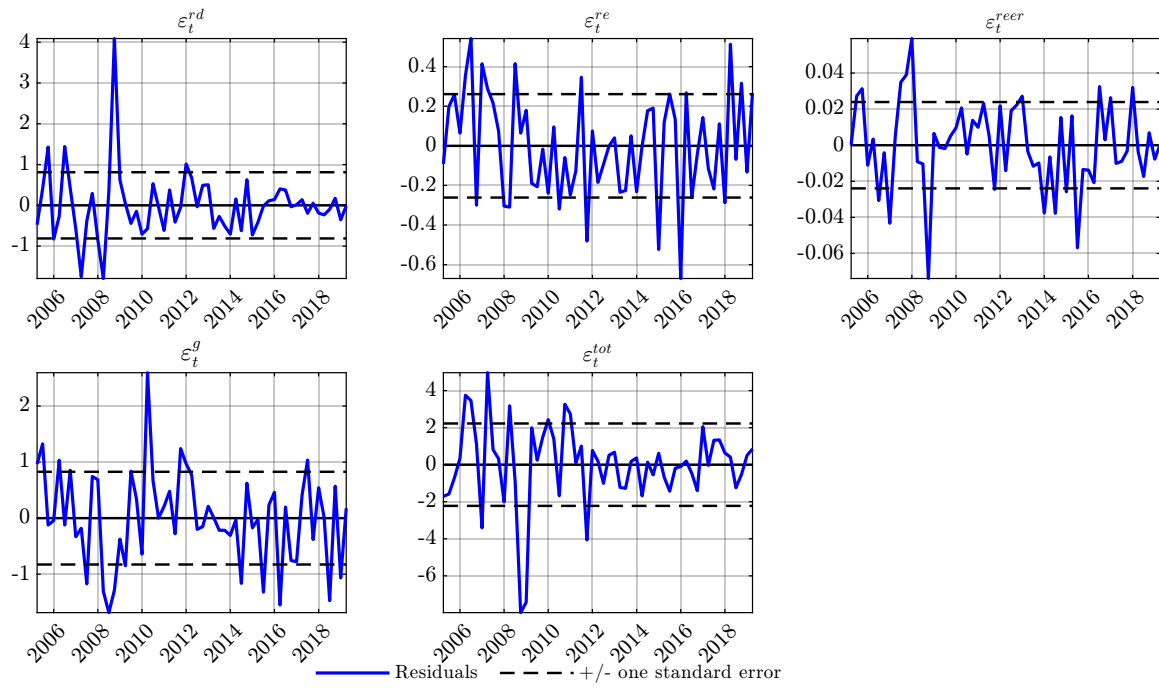


Figure A2.2: Residuals

Stability condition

Eigenvalue moduli
0.726557
0.703395
0.703395
0.428727
0.059754

Optimal lag selection by information criteria

Lag	AIC	HQIC	SBIC
0	7.27865	7.35058	7.46627
1	4.80094	5.23251*	5.92665*
2	4.86574	5.65695	6.92955
3	5.28376	6.43462	8.28567
4	4.85801	6.36852	8.79802
5	4.68362*	6.55377	9.56173

Portmanteau white-noise test on residuals

Variable	Q statistic	prob > $\chi^2(1)$
Domestic interest rate	1.5892	0.2074
Foreign interest rate	1.9923	0.1581
Log REER	0.6278	0.4282
Growth	0.1351	0.7132
ToT index	1.8479	0.1740

Annex 3: Multicollinearity

The original specifications following Ghosh et al. (2013) and Everaert and Jansen (2018) (specifications (1) and (2), respectively) are shown below; they suffer from multicollinearity problems, as indicated by their variance inflation factor. This problem occurs with all possible estimation methods (not only *Difference-GMM*) and is even amplified for some of them. One possible remedy is to eliminate one of the variables, as specifications (2) and (3) do; as specification (3) shows, the sign changes and the significance observed in (1) disappears, which would reflect the multicollinearity between the variables. An alternative solution is to combine variables through division, for example dividing the population of dependents over 64 by the population of dependents under 15; in this way, one can measure how aged the dependent population is and whether this will mean an additional demographic burden or, conversely, a demographic dividend. Another possible remedy is to change the variables, which is what this paper proposes by using $adr_{i,t}$ together with $adrol_{i,t+20}$ (not shown here, but it reaches a variance inflation factor of 3.12).

	(1) Diff-GMM	(2) Sys-GMM	(3) FE	(4) FE2	(5) MG
$pb_{i,t-1}$	0.5386*** (0.0023)	0.5868*** (0.0033)	0.5343*** (0.0034)	0.5411*** (0.0022)	0.5659*** (0.0036)
$d_{i,t-1}$	0.0323*** (0.0005)	0.0303*** (0.0005)	0.0326*** (0.0005)	0.0323*** (0.0004)	0.0312*** (0.0005)
$ygap_{i,t}$	0.1477*** (0.0027)	0.1436*** (0.0029)	0.1475*** (0.0019)	0.1479*** (0.0029)	0.1426*** (0.0025)
$totgap_{i,t}D_{exp}$	0.5362*** (0.0127)	0.5347*** (0.0114)	0.5146*** (0.0130)	0.5254*** (0.0103)	0.5313*** (0.0177)
$adr_{i,t}$	0.0232*** (0.0063)			-0.0001 (0.0045)	-0.0261*** (0.0058)
$adr_{i,t+20}$	-0.0730*** (0.0063)		-0.0691*** (0.0048)		
$adrolold_{i,t}$		-0.2881*** (0.0333)			
$adrolold_{i,t+20}$		0.0168 (0.0212)			
$young/old_{i,t+20}$					-3.9882*** (0.1981)
Constant	0.9167*** (0.2687)	1.9167*** (0.0788)	2.1208*** (0.2845)	-1.8947*** (0.3028)	1.5376*** (0.3901)
Mean squared error	3.19	3.33	3.18	3.20	3.27
FE Chile	1.7436	1.5806	1.2425	2.1261	1.2728
VIF adr	30.88	30.60	3.13	2.74	2.84

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

(1), (2) and (3) use robust standard errors; (4) clusters by country

Annex 4: Fiscal reaction function estimates, reduced sample and additional controls

Estimating from 1990 with a subsample of middle-income countries: Chile, Argentina, Uruguay, Colombia, Peru, Venezuela, Mexico, Costa Rica, Dominican Republic, Panama, Turkey, Romania, Poland, Hungary, Croatia, Bulgaria, South Africa, Malaysia, and Thailand.

Table 4: Fiscal reaction function estimates, reduced sample

	(1) Diff-GMM	(2) Sys-GMM	(3) FE	(4) FE2	(5) MG
$pb_{i,t-1}$	0.4456*** (0.0501)	0.6205*** (0.0361)	0.6114*** (0.0370)	0.6198*** (0.0373)	0.5715 (0.9003)
$d_{i,t-1}$	0.0562*** (0.0076)	0.0271*** (0.0050)	0.0286*** (0.0060)	0.0657 (0.0336)	0.0317*** (0.0082)
$ygap_{i,t}$	0.1712*** (0.0218)	0.1140*** (0.0218)	0.0659* (0.0312)	0.0641* (0.0311)	0.0911 (0.1710)
$adr_{i,t}$	-0.0961** (0.0330)	-0.0055 (0.0193)	-0.0044 (0.0212)	-0.0081 (0.0213)	0.0034 (0.2634)
$adrol d_{i,t+20}$	-0.1949*** (0.0287)	-0.0650*** (0.0150)	-0.1001*** (0.0222)	-0.1107*** (0.0230)	-0.1299 (0.4894)
$totgap_{i,t} D_{exp}$	0.8364*** (0.1629)	1.2134*** (0.2312)	0.5407*** (0.1383)	0.5467*** (0.1383)	0.5427 (1.1183)
$d_{i,t-1}^2$				-0.0004 (0.0006)	
$d_{i,t-1}^3$				0.0000 (0.0000)	
Constant	6.9647** (2.2316)	0.4178 (1.2530)	1.1132 (1.4893)	0.7667 (1.5262)	1.2327** (0.4139)
Observations	473	489	476	476	476
Countries	17	17	17	17	17
Mean squared error	2.00	2.38	3.01	3.00	2.71
F statistic	60	101	85	64	18
Instruments	123	177			
Residual variance Chile	3.80	6.50	3.53	3.72	3.06
FE Chile	1.6568	1.0152	1.0616	1.3737	-0.1741
$d_{i,t-1}$ Chile					0.0372
d_{ss} Low scenario	74.3403	40.0026	53.2290	53.9205	78.6721
d_{ss} Middle scenario	109.8166	56.0623	76.1410	87.1606	101.0498
d_{ss} High scenario	177.7607	87.0115	120.0220	136.9702	143.9074

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

(1), (2) and (3) use robust standard errors; (4) clusters by country

Table 5: Fiscal reaction function estimates with additional controls

Estimation of the fiscal reaction function including controls for annual *inflation* (WEO) and institutional variables from the Worldwide Governance Indicators (*Regulatory Quality* and *Political Stability*), available at <http://info.worldbank.org/governance/wgi/>.

	(1) Diff-GMM	(2) Sys-GMM	(3) FE	(4) FE2	(5) MG
$L.pb_{i,t-1}$	0.4330*** (0.0041)	0.5982*** (0.0050)	0.5885*** (0.0173)	0.5870*** (0.0172)	0.5278 (1.8428)
$d_{i,t-1}$	0.0369*** (0.0013)	0.0242*** (0.0006)	0.0197*** (0.0023)	0.0405*** (0.0110)	0.0500*** (0.0059)
$ygap_{i,t}$	0.2210*** (0.0052)	0.1763*** (0.0050)	0.0352 (0.0189)	0.0394* (0.0189)	0.1358 (0.9279)
$adr_{i,t}$	-0.0185*** (0.0051)	-0.0441*** (0.0042)	0.0038 (0.0093)	0.0044 (0.0093)	-0.0027 (0.8156)
$adrol d_{i,t+20}$	-0.1468*** (0.0047)	-0.1337*** (0.0023)	-0.1244*** (0.0110)	-0.1256*** (0.0110)	-0.2022 (1.5054)
$totgap_{i,t} D_{exp}$	0.6049*** (0.0111)	0.4424*** (0.0282)	0.5125*** (0.0491)	0.5071*** (0.0491)	0.5614 (3.1660)
$RegQual_{i,t}$	-0.9846*** (0.1100)	1.0986*** (0.0969)	0.0772 (0.2356)	0.1096 (0.2380)	-0.3499 (24.3272)
$Inflation_{i,t}$	0.0000 (0.0001)	0.0028*** (0.0004)	0.0003 (0.0018)	0.0005 (0.0018)	0.0009 (0.1826)
$PolStab_{i,t}$	-0.0390 (0.0720)	0.4889*** (0.0502)	0.1351 (0.1372)	0.1623 (0.1373)	0.1193 (34.4703)
$d_{i,t-1}^2$				-0.0001 (0.0001)	
$d_{i,t-1}^3$				0.0000 (0.0000)	
Constant	2.2307*** (0.3997)	3.6449*** (0.2760)	1.0397 (0.6252)	0.3826 (0.6702)	1.6688*** (0.3280)
Observations	2332	2436	2434	2434	2434
Countries	105	105	105	105	105
Mean squared error	2.89	3.41	4.25	4.22	3.58
F statistic	5744	7781	200	167	20
Instruments	106	149			
AR(1)		0.00			
AR(2)		0.17			
Sargan–Hansen p-value	0.31	0.99	0.00	0.00	
FE Chile					

Standard errors in parentheses

Annex 5: Extension with other capital requirements

A relevant extension to the model is the incorporation of other capital requirements s_t , which represent exceptional flows that are not considered within the primary balance but do impact debt. These flows are usually omitted since they do not represent very significant amounts, but they can still alter the results. In the Chilean case, these include recognition bonds, purchases of state-guaranteed student loan portfolios, and capitalization of state-owned enterprises with retained earnings, among others (Consejo Fiscal Autónomo, 2021). To add variation in this variable, it is incorporated as a stochastic variable $s_{t,k} \stackrel{\text{iid}}{\sim} \mathcal{N}(\bar{s}, \sigma_s^2)$, where $\bar{s}$ is its historical average of 0.4232 and σ_s the historical standard deviation of 0.7266¹¹. The new budget constraint becomes:

$$d_{t,k} = \frac{[\theta(1 + r_{t,k}^f)(1 + \Delta reer_{t,k}) + (1 - \theta)(1 + r_{t,k}^d)]}{(1 + g_{t,k})} d_{t-1,k} - pb_{t,k} + s_{t,k}$$

The comparison with the baseline case of fiscal reaction function specification (1) (*Difference-GMM*) is shown in Figure A5. The debt trajectory increases more, since other capital requirements raise debt on average. The primary balance trajectory is similar in both cases; as debt levels diverge, the presence of other capital requirements generates a slightly larger average primary balance response, due to the fiscal reaction function.

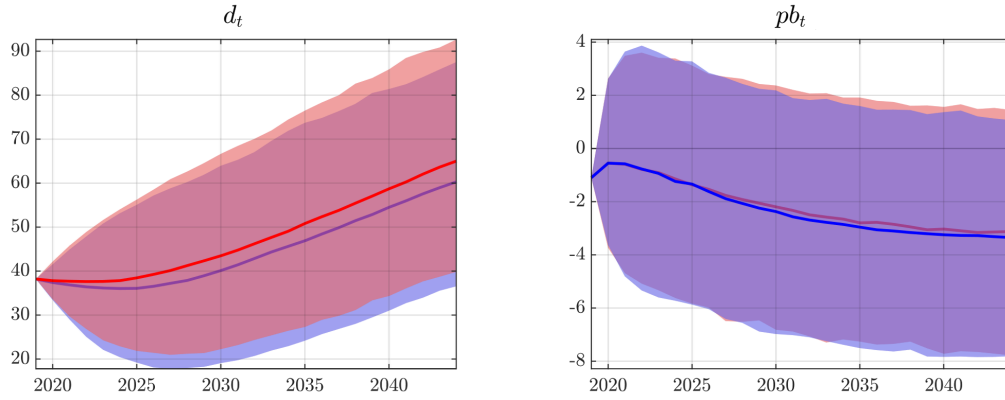


Figure A5: Fiscal trajectories with other capital requirements (red) and the baseline case (blue)

¹¹Data provided by the Autonomous Fiscal Council

Annex 6: Default risk extension

One way to address the criticism that in this model interest rates do not respond endogenously to higher indebtedness is the inclusion of a risk premium, which allows effects not only from non-fiscal variables to fiscal ones, but also lets fiscal variables such as the debt level affect the interest rates at which the State borrows. In particular, the proposed approach connects the *public debt level* with the *probability of default*, and these in turn with *interest rates*. Taking advantage of existing estimates by Méndez-Vizcaíno and Moreno-Arias (2021) of the response of the default probability to higher debt levels, a continuous function of this variable can be created through an Akima interpolation, as shown in Figure A6.1:

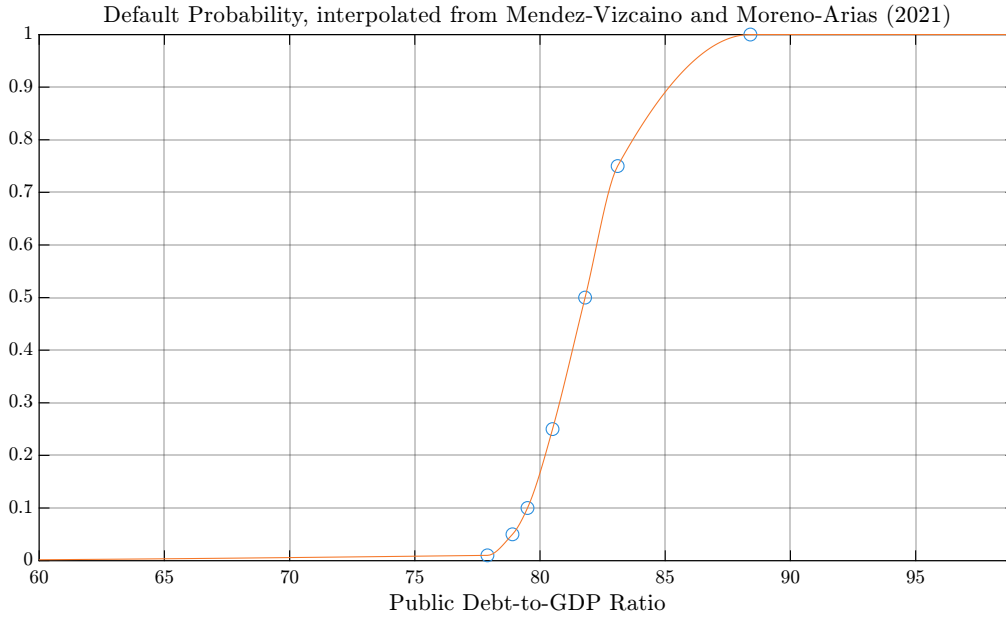


Figure A6.1: Default risk, Akima interpolation

Then, using the no-arbitrage equation following Fournier and Bétin (2018), the utility of a default-risk-free bond with rate r (as in the original model) must equal that of the risky bond with rate r^* :

$$U(1 + r) = (1 - p(d))U(1 + r^*) + p(d)U(\theta(1 + r^*))$$

where $p(d)$ represents the probability of default as a function of the debt level, and θ is the recovery rate, which represents how much of the total debt is recovered if a default occurs. Following Bi (2012), combining the declared default rates of emerging countries with the haircuts yields the effective default rates (which would be $1 - \theta$). Except for the cases of Argentina (0.41), Russia (0.185–0.233), and Ecuador (0.188), in which both the declared default rate and the *haircut* were high, the remaining cases do not exceed 10%:

Default rates in emerging countries

Country	Date	Declared default rate	Haircut	Effective default rate
Venezuela	Jul 1998	0,01	~ 0	~ 0
Rusia	Aug 1998	0,37	0,5-0,63	0,185-0,233
Ucrania	Sep 1998	0,09	0,3	0,027
Ucrania	Jan 2000	0,075	0,28	0,021
Pakistán	Jul 1999	0,035	0,35	0,0123
Ecuador	Ago 1999	0,46	0,4	0,188
Perú	Sep 2000	0,2	~ 0	~ 0
Argentina	Nov 2001	0,57	0,72	0,41
Uruguay	2003	0,53	0,13	0,07
Rep. Dominicana	2005	0,13	0,05	0,0065
Belice	2006	0,267	0,3	0,08
Average		0,248	0,281	0,093

If we assume a CRRA utility function $U(x) = \frac{x^{1-\alpha}}{1-\alpha}$, where α represents the risk aversion coefficient:

$$(1+r)^{1-\alpha} = (1-p(d))(1+r^*)^{1-\alpha} + p(d)\theta^{1-\alpha}(1+r^*)^{1-\alpha}$$

Writing $(1+r^*)$ as $(1+r)\phi$, where ϕ would represent how much the interest rate changes relative to the baseline case, we can simplify:

$$1 = (1-p(d))\phi^{1-\alpha} + p(d)\theta^{1-\alpha}\phi^{1-\alpha}$$

Thus, for example, if we assume $\alpha = 0$ (risk neutrality), solving for ϕ yields:

$$\phi(\alpha = 0, p(d)) = \frac{1}{1-p(d) + p(d)\theta}$$

If instead we assume risk aversion with $\alpha = 1$ (the utility function would be $U(x) = \ln(x)$), ϕ would be:

$$\phi(\alpha = 1, p(d)) = \theta^{-p(d)}$$

This ϕ can then be incorporated into the government budget constraint as an additional factor that increases the interest rate at which the State borrows in the trajectory simulations, as a function of the default probability at that time (given the debt level in that period):

$$d_{t,k} = \frac{\phi(\alpha, p(d_{t-1,k}))[\theta(1+r_{t,k}^f)(1+\Delta reer_{t,k}) + (1-\theta)(1+r_{t,k}^d)]}{(1+g_{t,k})}d_{t-1,k} - pb_{t,k}$$

Results of these projections based on fiscal reaction function specification (1) (*Difference-GMM*) are shown below. The projections are not altered relative to the baseline case (the median trajectory of the simulations is shown in red), except in the tails, where debt exceeds 80% and falls into explosive debt spirals. In this way, the trajectories cease to be symmetric around the median case; instead, there are additional risks of increasing debt excessively. The parameter α is not decisive for the trajectories, in contrast to θ , which does increase debt; the question would be what its correct level is. The method for incorporating default risk into

the model proves efficient as an extension of the baseline model to address endogeneity problems.

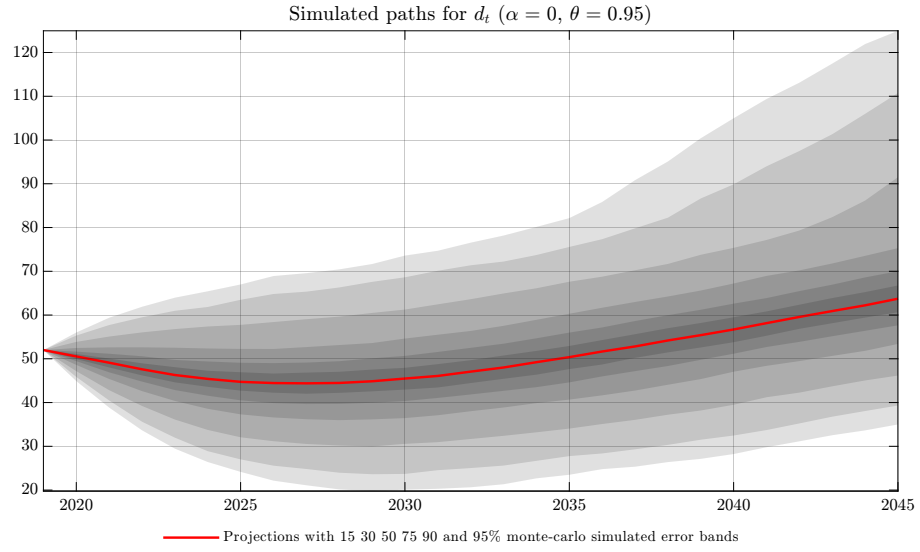


Figure A6.2: Projections with default risk incorporated: $\alpha = 0$ and $\theta = 0.95$

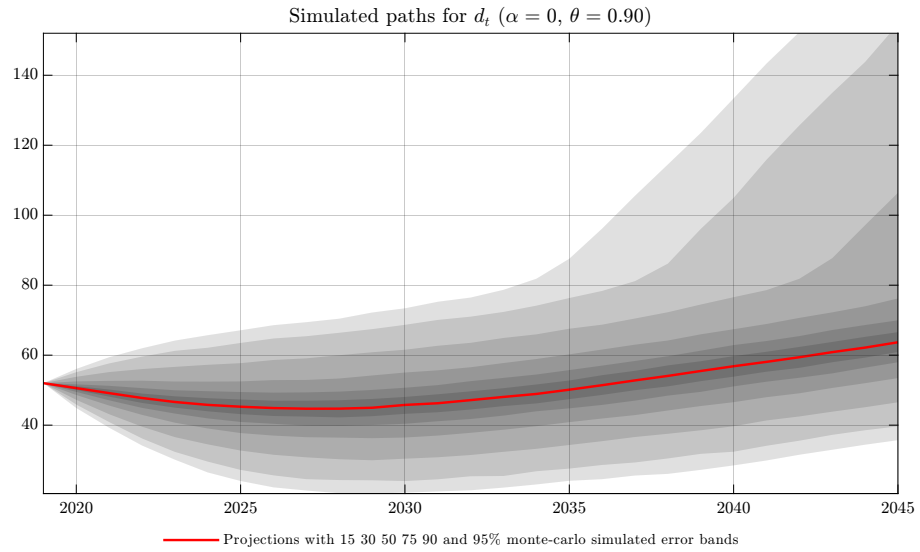


Figure A6.3: Projections with default risk incorporated: $\alpha = 0$ and $\theta = 0.9$

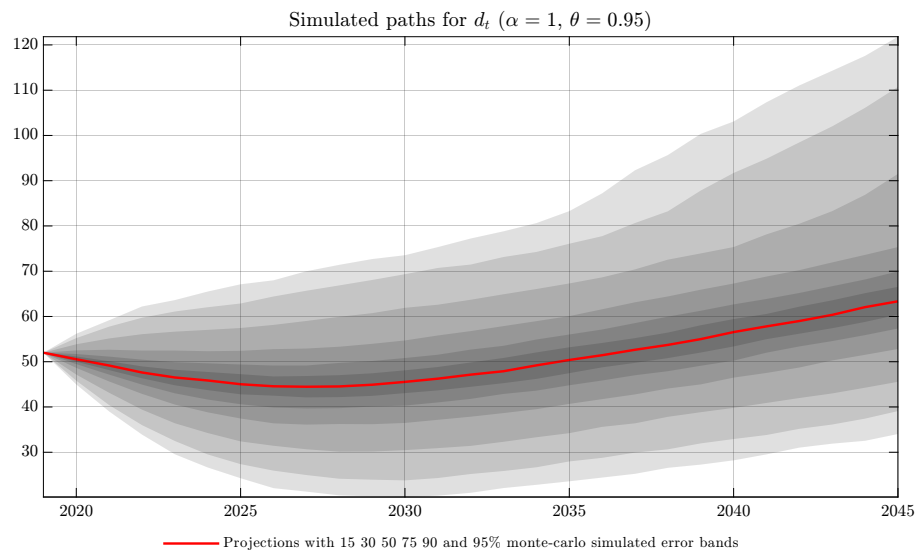


Figure A6.4: Projections with default risk incorporated: $\alpha = 1$ and $\theta = 0.95$

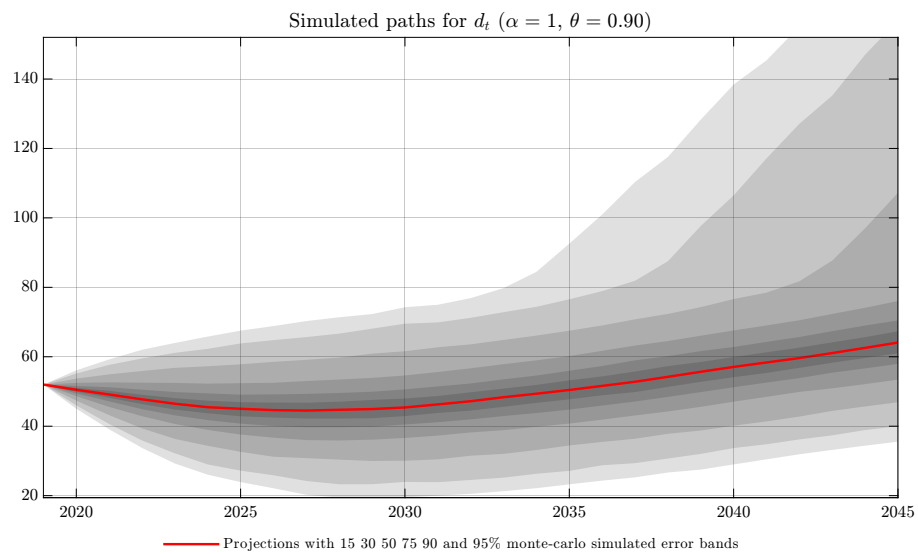


Figure A6.5: Projections with default risk incorporated: $\alpha = 1$ and $\theta = 0.9$

Annex 7: Deriving the Steady State

This annex aims to derive, analytically, the steady-state level of public debt in the proposed fiscal sustainability model. This requires establishing a relationship between the long-run debt level in the model and its parameters. Once this is done, this function can be used to estimate comparative statics.

The fiscal reaction function for Chile is given by the following reduced-form relationship:

$$pb_t = \alpha + \delta pb_{t-1} + \rho d_{t-1} + \gamma ygap_t + \tau totgap_t + \beta_{ADR} adr_t + \beta_{FADR} adrol d_{t+20} + \mu_{CL} + \varepsilon_t$$

Meanwhile, the government budget constraint is given by:

$$d_t = \frac{[\theta(1 + r_t^f)(1 + \Delta reer_t) + (1 - \theta)(1 + r_t^d)]}{(1 + g_t)} d_{t-1} - pb_t$$

In the steady state, $\varepsilon_t = 0$ since the shocks would disappear. $ygap_t = 0$ and $totgap_t = 0$ because these variables stabilize at their steady-state levels and the gaps become zero. The same holds for $\Delta reer_t = 0$. The equations then become:

$$pb_{ss} = \alpha + \delta pb_{ss} + \rho d_{ss} + \beta_{ADR} adr_{ss} + \beta_{FADR} adrol d_{ss} + \mu_{CL}$$

$$d_{ss} = \frac{[\theta(1 + r_{ss}^f) + (1 - \theta)(1 + r_{ss}^d)]}{(1 + g_{ss})} d_{ss} - pb_{ss}$$

To simplify, we define $r_{ss} = \theta(1 + r_{ss}^f) + (1 - \theta)(1 + r_{ss}^d) - 1$, and expanding the equations yields:

$$pb_{ss} = \frac{\alpha + \delta pb_{ss} + \rho d_{ss} + \beta_{ADR} adr_{ss} + \beta_{FADR} adrol d_{ss} + \mu_{CL}}{1 - \delta}$$

$$pb_{ss} = d_{ss} \frac{r_{ss} - g_{ss}}{1 + g_{ss}}$$

Combining both solved equations gives:

$$\frac{\alpha + \delta pb_{ss} + \rho d_{ss} + \beta_{ADR} adrol d_{ss} + \beta_{FADR} adr_{ss} + \mu_{CL}}{1 - \delta} = d_{ss} \frac{r_{ss} - g_{ss}}{1 + g_{ss}}$$

$$\alpha + \beta_{ADR} adr_{ss} + \beta_{FADR} adrol d_{ss} + \mu_{CL} = d_{ss} (1 - \delta) \frac{r_{ss} - g_{ss}}{1 + g_{ss}} - \rho d_{ss}$$

Arriving at the steady-state debt level d_{ss} :

$$d_{ss} = \frac{(1 + g_{ss})[\alpha + \beta_{ADR} adr_{ss} + \beta_{FADR} adrol d_{ss} + \mu_{CL}]}{(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})}$$

This debt level coincides (within a certain margin of error) with what results from projecting the model over long time horizons until convergence, which validates the approach (simpler than estimating the entire model each time). It is then important to see which variables are most decisive for the steady-state debt level. The

partial derivatives with respect to the reaction function coefficients are:

$$\frac{\partial d_{ss}}{\partial \delta} = -\frac{(1 + g_{ss})[\alpha + \beta_{ADR}adr_{ss} + \beta_{FADR}adrol_{ss} + \mu_{CL}]}{[(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})]^2}(g_{ss} - r_{ss})$$

$$\frac{\partial d_{ss}}{\partial \delta} = d_{ss}^2 \frac{r_{ss} - g_{ss}}{(1 + g_{ss})[\alpha + \beta_{ADR}adr_{ss} + \beta_{FADR}adrol_{ss} + \mu_{CL}]}$$

In this case (given the estimated parameter values), the derivative would be positive, indicating that the greater the persistence of the primary balance, the higher the steady-state debt; moreover, it would be non-linear in the debt level.

$$\frac{\partial d_{ss}}{\partial \rho} = \frac{(1 + g_{ss})[\alpha + \beta_{ADR}adr_{ss} + \beta_{FADR}adrol_{ss} + \mu_{CL}]}{[(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})]^2}(1 + g_{ss})$$

$$\frac{\partial d_{ss}}{\partial \rho} = \frac{d_{ss}^2}{\alpha + \beta_{ADR}adr_{ss} + \beta_{FADR}adrol_{ss} + \mu_{CL}}$$

Given the estimated parameters, the derivative would be negative: the stronger the response to lagged debt, the lower the steady-state debt, since the response comes through higher primary balances.

$$\frac{\partial d_{ss}}{\partial \gamma} = 0$$

$$\frac{\partial d_{ss}}{\partial \tau} = 0$$

Naturally, the coefficients associated with the output gap and the terms of trade gap have no effect on steady-state debt.

$$\frac{\partial d_{ss}}{\partial \mu_{CL}} = \frac{1 + g_{ss}}{(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})}$$

The derivative with respect to the fixed effect is constant, independent of the debt level. It has a negative sign, indicating that the larger the fixed effect, the lower the steady-state debt, because the response will be a higher primary balance on average.

$$\frac{\partial d_{ss}}{\partial \beta_{ADR}} = adr_{ss} \frac{1 + g_{ss}}{(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})}$$

$$\frac{\partial d_{ss}}{\partial \beta_{FADR}} = adrol_{ss} \frac{1 + g_{ss}}{(1 - \delta)(r_{ss} - g_{ss}) - \rho(1 + g_{ss})}$$

The derivative with respect to the dependency ratio terms is negative and is amplified as dependency ratios increase, indicating that if the demographic transition weighs more on the primary balance, debt will rise.

Below is a comparative statics exercise to examine how decisive these fiscal reaction function coefficients are for the steady-state convergence level, altering one coefficient around its estimated value (by 0.1) without changing the rest of the model. Specification (1) of the reaction function (*Difference-GMM*) and the 2050

stabilization level of the dependency ratios are used.

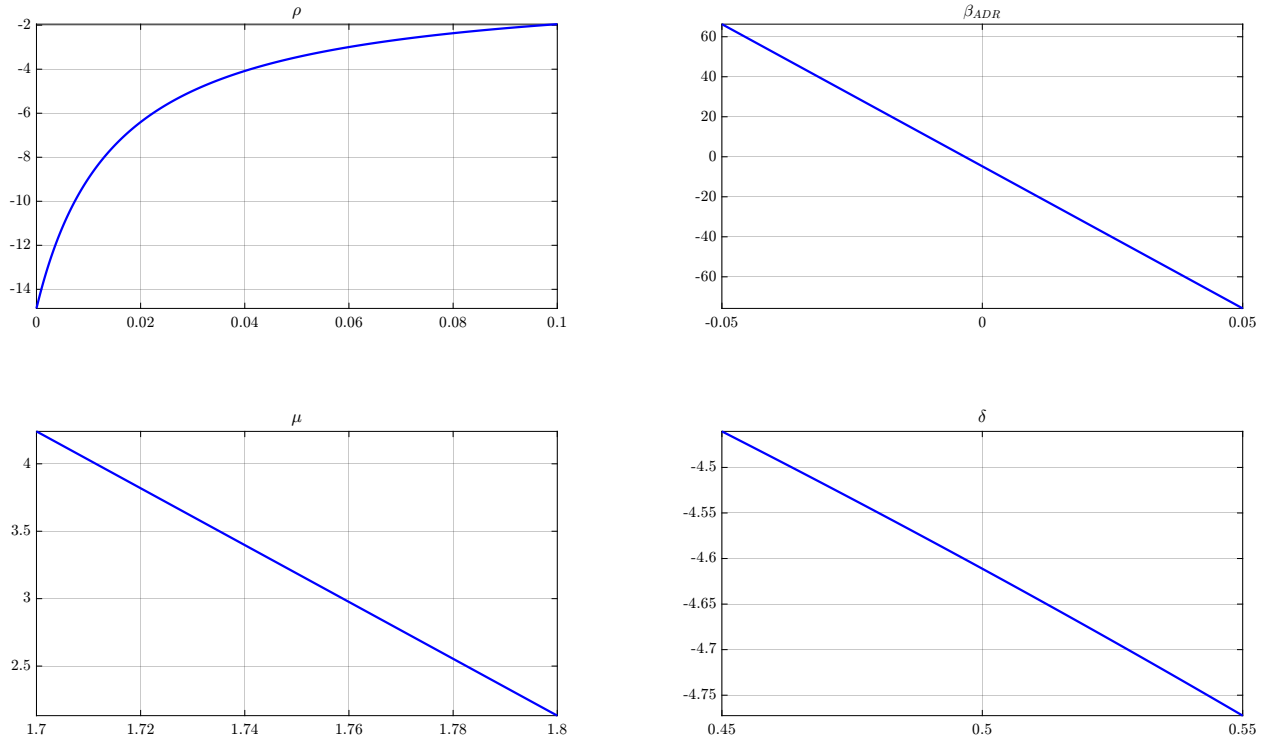


Figure A7.1: Comparative statics

The results are in line with what is expected from the estimated partial derivatives. β_{ADR} turns out to be highly decisive for the equilibrium obtained, as is ρ , which also has a highly non-linear impact, while μ_{CL} and δ have smaller effects.